PROBLEM 48

Two Connected Particles, One on Paraboloid

Original statement · Problems.pdf, p. 33

Multi-Particle Manifold

Two heavy particles P and Q, each of mass m, are connected with an inextensible, massless string of length ℓ as shown in the figure. Particle P is constrained to stay on a paraboloid, given implicitly by equation $z = cR^2$, where c is a known constant and we have introduced the usual cylindrical coordinates $\{R, \theta, z\}$. The string passes through a small hole at the origin O, and particle Q is restricted to hang along the vertical axis. The reference frame $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial. There is no friction in this problem. Assume that the string is long enough so that Q never reaches O and that it is always tense.

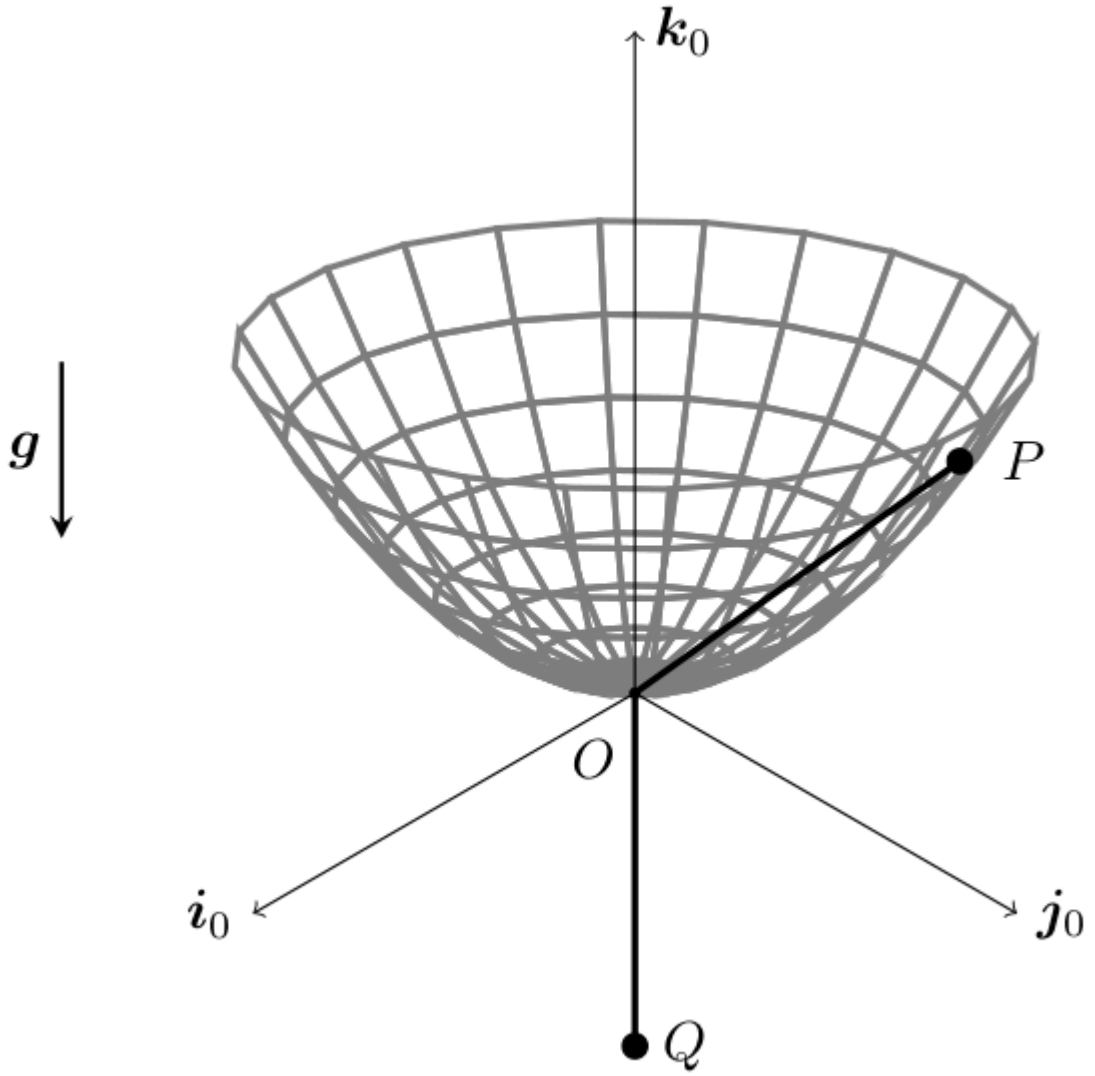


Figure: Paraboloid with P on its surface, string through the hole at O and Q hanging along the vertical axis k0 (Problems.pdf, p. 33).

(a) How many effective degrees of freedom does the system have? justify your answer.

(b) Give the expressions for the position, velocity, and acceleration of P and Q with respect to S_0 as a function of R, θ and their derivatives. If you use any vector basis other than $\{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ to write your solution, define it first.

(c) Find a normal vector and two linearly independent tangent vectors to the paraboloid as a function of R, θ. They do not need to be unitary.

(d) Identify and write down the general expression of all the forces acting on P and Q, introducing new unknowns as needed.

(e) Find the condition that the solution $R(t)$ must satisfy so that the string is indeed always in tension.

(f) Using S_0 for your analysis, write down two second-order differential equations for R and θ, that do not depend on any other unknowns.

(g) Find two scalar conservation laws (i.e., first integrals).

(h) Assuming that initially $R = 1/c, \theta = 0, \dot{R} = 0, \dot{\theta} = d$, where $d \neq 0$ is a given value, write down an equation in the form $f(R) = 0$ whose roots determine the minimum and maximum values of R in the subsequent motion. You do not need to solve this equation.

Worked Solution (4-Phase Methodology)

`PHASE 1`

Constraints, Degrees of Freedom & Coordinates

- **System:** two heavy particles P and Q of mass m are joined by an inextensible massless cord of length ℓ that passes through a small hole at the origin O. P moves without friction on the paraboloid $z = cR^2$ ($c > 0$, cylindrical coordinates $\{R, \theta, z\}$); Q is restricted to hang along the vertical axis Oz. The frame S_0 is inertial, there is no friction, the cord is always taut and Q never reaches O.
- **Constraints (a):** two free particles have 6 coordinates, and there are 4 independent constraints:
  - P on the paraboloid: $z_P = cR^2$ (1; bilateral, since the statement says P is constrained to stay on the paraboloid).
  - Q on the axis: $x_Q = y_Q = 0$ (2).
  - Constant cord length: $|\overrightarrow{OP}| + |z_Q| = \ell$ (1; unilateral, $T \geq 0$).
- **Degrees of freedom:** $\mathrm{CDOF} = 6 - 4 = 2$, with (R, θ).

- **Initial state (h):** $R(0) = 1/c, \theta(0) = 0, \dot{R}(0) = 0, \dot{\theta}(0) = d \neq 0$.
- **Parameters:** m, c [m^{-1}], ℓ, g, d [s^{-1}].

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$.
- **Cylindrical basis:** $\mathcal{B}_1 = \{\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{k}\}$ with $\mathbf{k} = \mathbf{k}_0$, the inertial basis rotated by $\theta(t)$ about $\mathbf{k}$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

Multiplying out, RR^T has diagonal entries $\cos^2 + \sin^2 = 1$ of the rotation angle (and 1) and off-diagonal entries $\pm(\cos\sin - \sin\cos) = 0$, so it is the identity. with $\dot{\mathbf{e}}_R = \dot{\theta}\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot{\theta}\,\mathbf{e}_R$.

- **Surface-adapted basis.** With $s = \sqrt{1 + 4c^2R^2}$, the unit meridian tangent is $\mathbf{e}_g = (\mathbf{e}_R + 2cR\,\mathbf{k})/s$, the unit normal pointing into the bowl is $\hat{\mathbf{n}} = (-2cR\,\mathbf{e}_R + \mathbf{k})/s$, and $\mathbf{e}_g \times \mathbf{e}_\theta = \hat{\mathbf{n}}$, so

$$[{}_1R_2] = \begin{pmatrix} 1/s & 0 & -2cR/s \\ 0 & 1 & 0 \\ 2cR/s & 0 & 1/s \end{pmatrix}, \qquad \det[{}_1R_2] = \frac{1 + 4c^2R^2}{s^2} = 1, \qquad [{}_1R_2][{}_1R_2]^T = I$$

PHASE 3

Derivation

3.1 Kinematics

(b) Position, velocity and acceleration

Pedagogical Justification: the cord runs straight from O to P, so the length of its part on the side of P is $\rho(R) = |\mathbf{r}_P| = R\sqrt{1 + c^2R^2}$, and Q hangs at the depth $\ell - \rho$:

$$\mathbf{r}_P = R\,\mathbf{e}_R + cR^2\,\mathbf{k}, \qquad \mathbf{r}_Q = -\left(\ell - \rho(R)\right)\mathbf{k}$$

- **Derivatives of ρ** (chain rule, with $w = \sqrt{1 + c^2R^2}$): $\rho' = \dfrac{d\rho}{dR} = w + \dfrac{c^2R^2}{w} = \dfrac{1 + 2c^2R^2}{w}$ and, by the quotient rule with $w' = c^2R/w$, $\rho'' = \dfrac{4c^2R\,w - \left(1 + 2c^2R^2\right)c^2R/w}{w^2} = \dfrac{c^2R\left[4\left(1 + c^2R^2\right) - \left(1 + 2c^2R^2\right)\right]}{w^3} = \dfrac{c^2R\left(3 + 2c^2R^2\right)}{w^3}$.
- **Particle P** (product rule and Poisson):

$$\mathbf{v}_P = \dot{R}\,\mathbf{e}_R + R\dot{\theta}\,\mathbf{e}_\theta + 2cR\dot{R}\,\mathbf{k}, \qquad v_P^2 = \dot{R}^2\left(1 + 4c^2R^2\right) + R^2\dot{\theta}^2$$

$$\mathbf{a}_P = \left(\ddot{R} - R\dot{\theta}^2\right)\mathbf{e}_R + \left(R\ddot{\theta} + 2\dot{R}\dot{\theta}\right)\mathbf{e}_\theta + 2c\left(\dot{R}^2 + R\ddot{R}\right)\mathbf{k}$$

(Notes Sec. 2.6.2: product rule on $\dot{R}\,\mathbf{e}_R$ and $R\dot{\theta}\,\mathbf{e}_\theta$ with $\dot{\mathbf{e}}_R = \dot{\theta}\,\mathbf{e}_\theta$, $\dot{\mathbf{e}}_\theta = -\dot{\theta}\,\mathbf{e}_R$, and $\frac{d}{dt}\left(2cR\dot{R}\right) = 2c\left(\dot{R}^2 + R\ddot{R}\right)$.)

- **Particle Q:** $\mathbf{v}_Q = \rho'\dot{R}\,\mathbf{k}$ and $\mathbf{a}_Q = \left(\rho'\ddot{R} + \rho''\dot{R}^2\right)\mathbf{k}$.

(c) Tangent and normal vectors to the paraboloid

$$\mathbf{t}_R = \frac{\partial \mathbf{r}_P}{\partial R} = \mathbf{e}_R + 2cR\,\mathbf{k}, \qquad \mathbf{t}_\theta = \frac{\partial \mathbf{r}_P}{\partial \theta} = R\,\mathbf{e}_\theta, \qquad \mathbf{n} = \mathbf{t}_R \times \mathbf{t}_\theta = R\left(-2cR\,\mathbf{e}_R + \mathbf{k}\right)$$

3.2 Forces

(d) Forces on P and Q

- **On P:** its weight $-mg\,\mathbf{k}$; the reaction of the smooth paraboloid $\mathbf{N} = N\,\hat{\mathbf{n}}$ (unknown magnitude N, normal to the surface); and the tension of the cord, directed toward the hole, $\mathbf{T}_P = -T\,\dfrac{\mathbf{r}_P}{|\mathbf{r}_P|}$.
- **On Q:** its weight $-mg\,\mathbf{k}$, the tension $\mathbf{T}_Q = +T\,\mathbf{k}$ (same cord, smooth hole), and the horizontal reaction $\mathbf{R}_Q$ of the vertical guide.
- **Inertia forces:** none (S_0 is inertial).

3.3 Equations of Motion

(f) Two second-order equations for R and θ

> **Pedagogical Justification:** the unknowns are R, θ, T, N and the two components of $\mathbf{R}_Q$, i.e. 6 unknowns, and Newton's second law for P and Q gives 6 scalar equations. Projecting the equation of P on the tangent vectors removes the normal reaction N, and the equation of Q gives T and $\mathbf{R}_Q = \mathbf{0}$ (since $\mathbf{a}_Q \parallel \mathbf{k}$):

$$Q,\ \mathbf{k}:\ m\left(\rho'\ddot{R} + \rho''\dot{R}^2\right) = T - mg \implies T = m\left(g + \rho'\ddot{R} + \rho''\dot{R}^2\right)$$

- **Equation for θ.** Projecting on $\mathbf{e}_\theta$ (no component of the reaction, of gravity or of the tension, which lies in the meridian plane): $m\left(R\ddot{\theta} + 2\dot{R}\dot{\theta}\right) = 0$, i.e. $\dfrac{m}{R}\dfrac{d}{dt}\left(R^2\dot{\theta}\right) = 0$.
- **Equation for R.** Projecting on $\mathbf{t}_R$: the tension component is $-T\,\dfrac{\mathbf{r}_P}{|\mathbf{r}_P|}\cdot\mathbf{t}_R = -T\rho'$ (because $\dfrac{\mathbf{r}_P}{|\mathbf{r}_P|}\cdot\mathbf{t}_R = \dfrac{R + 2c^2R^3}{Rw} = \rho'$) and the weight gives $-mg\,\mathbf{k}\cdot\mathbf{t}_R = -2cmgR$:

$$m\left[\ddot{R} - R\dot{\theta}^2 + 2cR\cdot 2c\left(\dot{R}^2 + R\ddot{R}\right)\right] = -2cmgR - T\rho'$$

Substituting T from the equation of Q:

$$\boxed{\left(1 + 4c^2R^2 + \rho'^2\right)\ddot{R} + \left(4c^2R + \rho'\rho''\right)\dot{R}^2 - R\dot{\theta}^2 + 2cgR + g\rho' = 0}, \qquad \boxed{R\ddot{\theta} + 2\dot{R}\dot{\theta} = 0}$$

with $\rho'^2 = \dfrac{(1 + 2c^2R^2)^2}{1 + c^2R^2}$.

3.4 Integration and Answers

(e) Condition for the cord to remain in tension

> **Pedagogical Justification:** the cord can only pull, so $T > 0$ at all times. From the equation of Q:

$$T = m\,(g + \ddot{z}_Q) > 0 \iff \ddot{z}_Q = \rho'\ddot{R} + \rho''\dot{R}^2 > -g, \qquad \text{i.e.} \quad \frac{1 + 2c^2R^2}{\sqrt{1 + c^2R^2}}\ddot{R} + \frac{c^2R\left(3 + 2c^2R^2\right)}{\left(1 + c^2R^2\right)^{3/2}}\dot{R}^2 > -g$$

(g) Two scalar conservation laws (first integrals)

- **Azimuthal integral (first integral of the θ equation).** From the equation for θ and $a_\theta = \frac{1}{R}\frac{d}{dt}\left(R^2\dot{\theta}\right)$ (product rule: $\frac{1}{R}\left(2R\dot{R}\dot{\theta} + R^2\ddot{\theta}\right) = R\ddot{\theta} + 2\dot{R}\dot{\theta}$):

$$\boxed{R^2\dot{\theta} = \text{const} = R(0)^2\,d = \frac{d}{c^2}} \implies \dot{\theta} = \frac{d}{c^2R^2}$$

- **Mechanical energy.** The reaction of the paraboloid does no work ($\mathbf{N}\cdot\mathbf{v}_P = 0$), the tension does the work $-T\rho'\dot{R}$ on P and $+T\rho'\dot{R}$ on Q, which cancel (inextensible cord, smooth hole), the guide reaction is perpendicular to $\mathbf{v}_Q$, and gravity is conservative. With $V = mg\left(cR^2 + z_Q\right) = mg\left[cR^2 + \rho - \ell\right]$:

$$E = \tfrac{1}{2}m\left[\left(1 + 4c^2R^2 + \rho'^2\right)\dot{R}^2 + R^2\dot{\theta}^2\right] + mg\left(cR^2 + \rho\right) - mg\ell = \text{const}$$

(h) Equation $f(R) = 0$ for the extreme values of R

> **Pedagogical Justification:** at the extremes $\dot{R} = 0$, so only $\frac{1}{2}mR^2\dot{\theta}^2$ remains of the kinetic energy, with $R^2\dot{\theta}^2 = d^2/(c^4R^2)$ from the azimuthal integral. The constant $-mg\ell$ cancels in $E - E_0$. At $R(0) = 1/c$: $\rho = \sqrt{2}/c$ and $R^2\dot{\theta}^2 = d^2/c^2$. Per unit mass:

$$\boxed{f(R) = \frac{d^2}{2c^4R^2} + g\left(cR^2 + R\sqrt{1 + c^2R^2}\right) - \frac{d^2}{2c^2} - \frac{(1 + \sqrt{2})\,g}{c} = 0}$$

- The allowed radii are those with $f(R) \leq 0$ (so that $\dot{R}^2 \geq 0$), and $R = 1/c$ is a root.

PHASE 4

Interpretation & Checks

- **Limit $c \to 0$ (flat plane):** $\rho = R$, $\rho' = 1$, $\rho'' = 0$, and the equation for R becomes $2\ddot{R} - R\dot{\theta}^2 + g = 0$, that of two equal particles joined through a hole in a horizontal plane.
- **Bounded precessing motion:** the particle moves between two parallel circles $R_{\min}$ and $R_{\max}$, with the azimuthal angle monotonic ($\dot{\theta} = d/(c^2R^2)$ has the sign of d).
- **Dimensional check:** $[c] = \mathrm{m}^{-1}$, $[d] = \mathrm{s}^{-1}$; $[cR^2] = [\rho] = \mathrm{m}$ and cR is dimensionless; every term of f has units $\mathrm{m}^2/\mathrm{s}^2$.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 48 in its solution keys, so the results are verified by independent derivation. Sympy confirms ρ', ρ'', the projections of $\mathbf{a}_P$ on $\mathbf{t}_R$ and $\mathbf{e}_\theta$, and numerical integration of the two equations of (f) ($c = d = g = 1$) conserves E to 2×10^{-12}, keeps $T/m \geq 0.303$ and bounds R between 0.482807 and 1, the roots of f ($f(0.482807) = 0$ and $f(1) = 0$; $f < 0$ between them).

Source: Problems.pdf